The Many Apportionment Paradoxes of the 2020 Iowa Democratic Presidential Caucuses

David McCune

If you remember anything about the 2020 Iowa Democratic presidential caucuses (they occurred before the first Covid lockdowns, so you are to be forgiven if you don't), it is probably that the reporting of election results was massively hampered by technology issues. "An embarrassing vote-reporting mess in the Iowa Democratic caucuses means there is so far no winner in the crucial opening contest," reported a typical article on `cnn.com` [5]. Other news outlets ran stories with headlines like "How the Iowa caucuses fell apart and tarnished the vote" [2] and "On the trail: making sense of the mess in Iowa" [10]. Much to the frustration of voters and presidential candidates alike, the Iowa Democratic Party announced the official results an astounding 24 days after the election, meaning that the winner of this first electoral contest in the Democratic presidential primary wasn't known until after two other primary elections, in New Hampshire and Nevada, had already occurred.

What has gone unreported, until this article, is that the 2020 Iowa Democratic caucuses were remarkable in another way: they produced a treasure trove of apportionment paradoxes. In fact, from the mathematical viewpoint of apportionment theory, these Iowa caucuses may have been the most paradox-riddled election in the history of presidential primaries. This article describes the paradoxes exhibited in the caucuses and provides some historical context for these paradoxical results.

The Apportionment Problem

We begin with a brief introduction to the theory of apportionment. This branch of mathematics traces its origins to Article I, Section 2 of the United States Constitution, which states:

> Representatives [to the US House of Representatives] ... shall be apportioned among the several States ... according to their respective Numbers.

In other words, if a state has $p\%$ of the US population, then the state should receive $p\%$ of the seats in the US House of Representatives. A mathematical problem (an *apportionment problem*) arises because $p\%$ of the seats is almost never an integer, but a seat represents a human representative, and thus somehow we must transform fractional amounts of seats into integers. "But wait," you may be thinking, "transforming nonintegers into integers is a 'problem' that any third-grader can solve: just round to the nearest integer." More on that in a minute.

To show how apportionment problems can be solved, consider a small country with a population of 300,000 people and six states, where the states have populations as shown in the Population row of Table 1. Suppose this country's House of Representatives contains 40 seats, which must be apportioned to the states in proportion to state population in accordance with Article I, Section 2. A natural starting point for considering how to apportion the 40 seats is to calculate each state's *quota*, which is the proportion of the state's population multiplied by 40. (This is the number of seats each state would receive if fractional seats were allowed.) The state quotas are displayed in the Quota row of Table 1.

First, note that we cannot solve this country's apportionment problem simply by rounding. If we round each quota to the nearest integer, then we apportion 15, 10, 5, 4, 4, and 3 seats to states A, B, C, D, E, and F, respectively, distributing 41 seats in total (one more than the 40 available). A natural response to this overallocation is, why not simply accept the apportionment with 41 seats? There are two short answers, one legal and one mathematical. The legal answer is that often the number of seats is set by law, which is the case for the 435 seats in the US House. The mathematical answer is that if we change the number of seats to 41, then the quotas also change, perhaps necessitating further changes in House size. In this example, if we raise the number of seats to 41 and recalculate quotas, nearest integer rounding would again yield an overallocation of one seat, and thus we would need to increase the size of the House to 42. Of course, it is possible that this process eventually stabilizes (in this example, nearest integer rounding with 43 seats works out), but it is undesirable to change the total amount of representation due to accidents of nearest integer rounding. Thus the size of the House is fixed in apportionment theory.

How can the apportionment problem be solved if we don't use nearest integer rounding? We describe three separate approaches, each of which has been used for prior apportionments of the US House. Most of the information provided can be found in [1]; a nice introduction to this material written for a student audience can be found in [14].

After the first US census in 1790, Alexander Hamilton proposed the following method for allocating House seats. First, give each state a number of seats equal to the floor of its quota. In our example, this results in an initial allocation

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Table 1. Three possible apportionments in a small country with six states

State	*A*	*B*	*C*	*D*	*E*	*F*
Population	111,615	72,291	34,222	33,377	26,598	21,897
Quota	14.88	9.64	4.56	4.45	3.55	2.92
Hamilton	15	10	5	4	3	3
Jefferson	16	10	4	4	3	3
Huntington–Hill	15	9	5	4	4	3

of (14, 9, 4, 4, 3, 2). Since we have rounded the quotas down, unless the quotas are all integers we have some amount $R > 0$ of seats remaining. Hamilton stipulated that the R remaining seats be given to the R states with largest fractional part of their quotas. In our example, there are four seats remaining, which are given to states A, B, C, and F (see the Hamilton row of Table 1), resulting in a final apportionment of (15, 10, 5, 4, 3, 3).

Why might Hamilton's method be a "good" way to allocate House seats? Note that each state always receives either the floor or ceiling of its quota (the method is said to satisfy the *quota rule*), which satisfies an intuitive notion of fairness in the context of proportional allocation. The method is not without its flaws, however: it is susceptible to many apportionment paradoxes, some of which we explore below. Hamilton's method was approved by Congress for the country's first apportionment, but use of the method was vetoed by George Washington in favor of a method created by Thomas Jefferson. Hamilton's method was used sporadically throughout the nineteenth century, but because it can produce paradoxical outcomes, it has not been seriously considered for apportioning US House seats since the 1900 census.

The apportionment method created by Thomas Jefferson can be defined as a method that sequentially gives seats to states one seat at a time. The basic idea is that if we have already handed out n seats, then the $(n+1)$st seat is given to the state in which the $(n+1)$st representative can represent the most people. In our example, the first seat goes to state A, because that representative would represent 111,615 people, more than in any other state (in this method, the first seat always goes to the most populous state). The second seat does not go to state A, because each representative would then represent 55,807.5 people; instead, the second seat goes to state B, so that the second representative represents 72,291 people. The third seat goes to state A, because each representative from A represents 55,807.5 people, whereas if the third seat went to B (respectively C), then this third representative would represent only 36,145.5 (respectively 33,377) people. The method continues in this fashion until all seats have been allocated. The resulting apportionment is shown in the Jefferson row of Table 1.

The advantage of Jefferson's method is that it avoids most of the apportionment paradoxes that we discuss in this article. The disadvantage from a fairness perspective is that the method does not satisfy the quota rule. In our example, state A receives more than the ceiling of its quota, suggesting that A has been unfairly favored. Jefferson's method frequently "breaks quota" in this manner, and this breaking of quota tends to favor the more populous states (as occurs in this example).[1] For this reason, the method was discarded for US House seat apportionment in the early nineteenth century.

The method that is currently used to apportion seats in the US House, the Huntington–Hill method, was invented by the mathematician Edward Huntington in 1921, based on a notion of fairness articulated by the statistician Joseph Hill in 1910. Hill thought that an apportionment method should ensure that "the ratios of the number of inhabitants per Representative should be nearly as uniform as possible." That is, the size of a congressional district in state X (which is the population of X divided by its number of seats) should be as close as possible to the size of a congressional district in state Y for any two states X and Y. If the goal of apportionment is fair representation, then Hill's fairness criterion makes complete sense. Of course, to make rigorous this notion of fairness we must define "as close as possible." Hill focused on the relative difference between the sizes of districts from two states; specifically, let DS_X (respectively DS_Y) be the size of a congressional district in state X (respectively Y). Hill stipulated that

$$\frac{\left|DS_X - DS_Y\right|}{\min(DS_X, DS_y)} \tag{1}$$

should be as small as possible for any pair of states, in the sense that any transfer of seats between the two states causes expression (1) to increase. Our running example shows that this fairness criterion is not satisfied by Hamilton's method: expression (1) evaluated for states B and E under Hamilton yields 0.226, but if one seat were transferred from B to E, then we would get the smaller value 0.208.

Huntington created a method that satisfies the above fairness criterion; under a Huntington–Hill apportionment, any transfer of seats between two states causes expression (1) to increase. Similar to Jefferson's method, the Huntington–Hill method can be framed in terms of sequential allocation, but in this case, the $(n+1)$st seat is given according

[1]If Jefferson's method were the only method ever used for US House seat apportionment, it would have given at least one large state more than the ceiling of its quota in every apportionment since 1860.

to a rule different from "represent the most people" (see [15] for a nice description of the definition). The resulting Huntington–Hill apportionment for our example is given in Table 1. Note that Huntington–Hill allocates nine seats to B and four to E, minimizing expression (1) with respect to seat transfers between B and E, in contrast to Hamilton.

Huntington–Hill satisfies an additional notion of fairness regarding the minimizing of differences between congressional district size across state lines. In our running example, the total population of the country is 300,000 and there are 40 seats, and thus the "ideal" size of a congressional district is $300000/40 = 7500$ people. Because of the nature of the apportionment problem, we cannot achieve district sizes of 7500 in each state. However, we can insist that there be as little variation from the ideal district size as possible. That is, we can allocate seats in such a way as to minimize the standard deviation of each state's district size from the ideal district size. The Huntington–Hill method always produces an apportionment that minimizes this standard deviation (see [6]). The reader can check that in our running example, the six state district sizes achieved by the Huntington–Hill apportionment have a smaller standard deviation from 7500 than the six district sizes achieved by Hamilton or Jefferson. This property of Huntington–Hill doesn't feature prominently in the debates over House seat apportionment, but it does play a role in other contexts. This is why, for example, the Democratic Party of Boulder County, Colorado, has used Huntington–Hill to allocate county delegates to precincts in the Colorado Democratic state caucuses.[2]

The Huntington–Hill method has been used for every US House apportionment since the 1940 census. The method has two main drawbacks, although usually these shortcomings are not considered to be nearly as serious as those of Hamilton's and Jefferson's methods. First, sometimes the method produces apportionments in which quota is broken for at least one state; however, this is a theoretically rare occurrence and has never happened in actual US House apportionments. Second, the method favors small states at the expense of the large (as can be seen in Table 1; note the apportionments for B and E under the three methods), but this bias is not nearly as pronounced as that of Jefferson's method, which strongly favors large states.

We have described three methods for solving the apportionment problem, but many others exist. For example, the methods of Jefferson and Huntington–Hill belong to a large class of sequential allocation methods called *divisor methods*, in which the allocation of the $(n+1)$st seat is based on different ways of evaluating which state is currently the most deserving; such methods are commonly used in parliamentary systems to allocate seats to political parties in proportion to vote share after a parliamentary election. Often, a method is created to satisfy a particular notion of fairness or to avoid undesirable outcomes (such as the paradoxes discussed below). Furthermore, the method chosen may depend on the context in which the apportionment problem arises. So far, we have focused on House seat apportionment; apportionment problems also arise in US presidential state primary elections, in which party delegates are apportioned to presidential candidates in proportion to vote share. In a setting in which delegates are apportioned according to candidates' electoral performance, our notions of fairness may change from the House seat context, and different methods may be appropriate. In fact, the methods of Huntington–Hill and Jefferson have never been used in presidential primaries, despite their prominence in debates over House seat apportionment. As we will see, Hamilton's method is used frequently in presidential primaries, and its paradoxical behavior is on full display in this context.

Apportioning Delegates in the Democratic Presidential Primary

We now describe the process by which the Democratic Party chooses its presidential nominee. For the sake of brevity we provide only those details that are necessary for this article; for a more complete description, see [3].

The Democratic presidential primary is a series of elections in all 50 states, the District of Columbia, and the American territories that is used to determine the Democratic Party nominee for president of the United States. During the months leading up to the Democratic National Convention in a presidential election year, each Democratic state[3] party holds a primary (or caucus[4]) election in which voters in the state vote for their preferred Democratic presidential candidate. Each state party has a set number of delegates that are up for grabs in its state primary election. Generally speaking, these delegates are apportioned to the candidates in proportion to the candidates' vote totals in the state election. As the primary season unfolds, each candidate accumulates delegates from the various state contests. Once all of the state primary elections have concluded, the candidate with a majority of delegates is named the presidential nominee at the national convention. If no candidate secures a majority of delegates, the Democratic Party has contingency rules to choose a nominee.

Each Democratic state party has three types of delegates to allocate in its state primary: district delegates, party-leaders-and-elected-officials (PLEO) delegates, and at-large delegates. The district delegates are divided among the state's congressional districts (CDs) and then apportioned based on election results in the individual districts, while the PLEO and at-large delegates are separately apportioned based on the statewide election results. For example, in 2020, the Iowa Democratic Party had 41 total delegates in its caucus election. Twenty-seven of these were district delegates and were thus distributed among Iowa's four CDs, with seven delegates going to CDs 1 and 2, eight delegates to CD 3, and five delegates to CD 4. Each district then had its own separate election to apportion its delegates to the presidential candidates in proportion to their vote totals in that

[2] Private communication.
[3] We use "state" as a catch-all term that includes DC and territories such as Puerto Rico.
[4] The differences between primary and caucus elections are not important in this article.

Table 2. Hamilton apportionment of the nine at-large delegates when there is no threshold (middle columns: Quota) and when the 15% threshold is applied (right columns: Adjusted Quota)

Candidate	SDEs	Quota q	$\lfloor q \rfloor$	Dels	Adj. Quota q	$\lfloor q \rfloor$	Dels
Buttigieg	562.9538	2.360	2	2	2.733	2	3
Sanders	562.0214	2.356	2	2	2.729	2	3
Warren	388.4403	1.628	1	2	1.886	1	2
Biden	340.3238	1.427	1	2	1.652	1	1
Klobuchar	263.8689	1.106	1	1	–	–	–
Others	29.4918	0.123	0	0	–	–	–
Total	2147.1000	9	7	9	9	6	9

CD. There were five PLEO and nine at-large delegates, each of which were separately apportioned based on the statewide election results. That is, instead of combining the 14 total statewide delegates into a single "statewide delegate" category, the five PLEO delegates were apportioned in a separate calculation from the nine at-large delegates, but each calculation used the same statewide election results.

Thus the 2020 Iowa caucuses consisted of six separate delegate allocation problems. The two statewide problems (PLEO and at-large) were solved based on the same statewide election results, and the four district problems were solved in each CD. How do state Democratic parties solve their delegate apportionment problems? The Democratic National Committee (DNC) stipulates that each state party must use Hamilton's method to apportion delegates to candidates in proportion to their vote totals. The Democratic delegate apportionment rules can be found in Rule 14 of the 2020 Delegate Selection Rules document, published by the DNC and available at `democrats.org`. Paraphrased, the rules state that delegates are to be apportioned in three steps:

1. Eliminate the candidates who receive less than 15% of the vote from the election, along with their votes.
2. Calculate each surviving candidate's quota, using only the votes of the surviving candidates.
3. Apportion delegates to the surviving candidates using Hamilton's method.

Thus, state Democratic parties use Hamilton's method to apportion delegates to candidates in proportion to vote totals, after applying a 15% viability threshold that eliminates weaker candidates (and their votes) from contention. These rules have been used at the statewide and CD level for every Democratic state primary since 1992, although the idea that state parties should use proportional allocation with a threshold dates back to the 1974 recommendations of the Mikulski Commission (see [7]).

The official rules from the DNC describe apportioning delegates based on vote totals. The Iowa Democratic Party complicates this by apportioning delegates based on what are called "state delegate equivalents" (SDEs). There are many well-written news articles that explain what an SDE is (see [12], for example); put simply, the candidate vote totals at each caucus precinct are translated into fractions of a state delegate, who attends the Iowa state Democratic convention, which occurs prior to the Democratic national convention. Each precinct has a set amount (often a fraction less than one) of SDEs available for allocation; the percentage of available SDEs earned by a candidate at a given precinct is approximately the candidate's precinct vote share. SDEs are essentially a stand-in for vote totals that the Iowa Democratic Party uses to shape its state convention. The use of SDEs (and the corresponding confusion surrounding what they are and why they exist in the first place) became another source of criticism for the Iowa Democratic Party after the 2020 caucuses. All of the delegate Hamilton math in Iowa is done using SDEs rather than vote totals, and thus we can simply take the SDEs as a given and run Hamilton's method on SDE totals.

Table 2 shows the Hamilton allocation of the nine at-large delegates using the Iowa SDE information, which was obtained from `thegreenpapers.com`. The SDEs column shows each candidate's statewide SDE total. The middle columns show a Hamilton apportionment for the nine at-large delegates without the 15% threshold. The quotas are obtained by dividing the given candidate's number of SDEs by 2147.1000 and then multiplying by 9. The resulting Hamilton apportionment for the top five candidates is (2, 2, 2, 2, 1), while all other candidates receive no at-large delegates. The last three columns show the apportionment that actually occurred when the 15% threshold was used. The adjusted quotas were calculated using a total of 1853.7393 SDEs, because that is how many SDEs remained after Klobuchar and the others were eliminated. Note that the use of a smaller denominator for total SDEs causes the quotas of the surviving candidates to increase after elimination of the weaker candidates. Once the new quotas are calculated, we apply Hamilton's method as before. The symbol – denotes that a candidate has fallen below the 15% threshold.

Table 2 shows the apportionment just for the nine at-large delegates. We will reference the apportionments for the other five buckets of delegates (PLEO and four CDs) below; Table 3 shows the complete SDE and apportionment data for candidates who received at least one delegate in the caucuses.

Table 3. The results of the 2020 Iowa Democratic Caucuses

Contest	Dels	Buttigieg		Sanders		Warren		Biden		Klobuchar	
		SDEs	Dels	SDEs	Dels	SDEs	Dels	SDEs	Dels	SDEs	Dels
CD 1	7	147.8163	2	145.2917	2	95.5602	1	102.7936	2	66.7423	–
CD 2	7	143.3223	2	157.1856	3	112.8221	2	72.7595	–	58.9573	–
CD 3	8	164.2389	3	159.2701	2	119.0656	2	92.8112	1	69.7616	–
CD 4	5	107.5763	2	100.2740	1	60.9924	–	71.959	1	68.4077	1
PLEO	5	562.9538	2	562.0214	1	388.4403	1	340.3238	1	–	–
at-large	9	562.9538	3	562.0214	3	388.4403	2	340.3238	1	–	–
Total	41		14		12		8		6		1

See www.thegreenpapers.com

Now that we understand the mathematics of delegate apportionment in the Iowa caucuses, we can discuss the panoply of paradoxes exhibited by this election.

Elimination Paradox

The first paradoxical outcome we examine occurred in apportioning the nine at-large delegates and can be observed in Table 2. Note that Klobuchar earned a delegate when there was no threshold, but her SDE count was not large enough to surpass the 15% threshold, and therefore her delegate needed to be reapportioned after she was eliminated. Thus it makes sense that one of the surviving candidates saw their delegate count increase after the threshold was applied. However, not one but two surviving candidates saw an increase in their apportionment, while one of the surviving candidates actually lost a delegate. When Hamilton's method is applied without the threshold, Biden earns two delegates, but when bottom candidates are eliminated and Hamilton's method is used on the adjusted quotas, he earns only one. Applying the 15% threshold cost Biden half of his at-large delegates. This outcome is paradoxical because thresholds are seemingly designed to redistribute delegates that were earned by weaker candidates. The threshold is meant to help, or at least not to harm, the "strong" or "good" candidates who survive elimination. If Biden earns two delegates at the statewide level before eliminating weaker candidates, why shouldn't he get to keep those two delegates after elimination? It is not clear whether Buttigieg or Sanders "took" Biden's delegate that he lost, but either way, the SDE counts among Buttigieg, Sanders, and Biden remain unchanged after the weaker candidates are eliminated, so why should either Buttigieg or Sanders gain a delegate at Biden's expense just because Klobuchar was eliminated?

This outcome is an example of an *elimination paradox*, which occurs when a surviving candidate's apportionment decreases as a result of eliminating weaker candidates who fall below some threshold. Historically, much of apportionment theory has been studied in the context of House seat allocation, and this paradox is of no relevance in that context because thresholds are never used. (Imagine if a state lost all of its representation in the House because its population share fell below some set percentage.) As a result, this paradox is not well studied. As far as I know, the only explorations of this paradox occur in [8] and [11]. The elimination paradox is of some relevance in a parliamentary context, in which seats are apportioned to political parties in proportion to their vote totals in an election. It is fairly common to have thresholds of varying size in these elections; however, the paradox hasn't been studied in this context either. It is shown in [8] that divisor methods such as those of Huntington–Hill and Jefferson are not susceptible to this paradox, and thus these methods have an advantage in this respect over Hamilton's method in the presidential primary setting.

How common are occurrences of the elimination paradox in Democratic presidential primaries? To answer this question, I obtained election data for the 2004, 2008, 2016, and 2020 Democratic presidential primaries, which was generously provided to me by Tony Roza, of thegreenpapers.com.[5] Roza has compiled nearly complete vote information at the statewide and CD level[6] for every state in these primaries, resulting in 1,981 elections for which I can check whether the elimination paradox occurred.[7] You may notice that the elimination paradox cannot occur in a two-candidate election; after I discard the two-candidate elections from the database, there are 1,882 remaining. Of these 1,882 elections, only 21 (1.1%) exhibited the elimination paradox. This paradox is a rare phenomenon in Democratic primaries.

Furthermore, in the Iowa at-large example of this paradox, one of the eliminated candidates earned a delegate before being eliminated. It is pointed out in [8] that we expect the paradox to occur less frequently if any of the

[5]Roza also has data for the 2000 and 2012 Democratic primaries, but these were essentially one-candidate contests and thus of no mathematical interest. I was unable to find any usable data for primaries prior to 2000.

[6]Sometimes the CD data must be estimated from county-level information provided by state parties or state secretary of state offices. I use the approximated data because it is the best I can do given the reluctance of state parties and state governments to share data at the district level.

[7]In this context, by "election" I mean a vote distribution along with a number of delegates to apportion. Every CD in every state is a separate election, along with the two statewide elections, one for PLEO and one for at-large apportionment (the number of PLEO delegates does not equal the number of at-large delegates). I created Python code to process the election data.

Table 4. The Alabama paradox for the at-large apportionment

Candidate	SDEs	Quota 8 dels	Dels	Quota 9 dels	Dels
Buttigieg	562.9538	2.429	2	2.733	3
Sanders	562.0214	2.425	2	2.729	3
Warren	388.4403	1.676	2	1.886	2
Biden	340.3238	1.469	2	1.652	1

The first apportionment is for eight delegates, and the second is for nine

eliminated candidates received delegates prior to elimination; much of the time, the paradox occurs when the eliminated candidates would have earned no delegates, and the apportionment of the surviving candidates is shuffled a bit after elimination. That is, in a "typical" example of an elimination paradox, all the candidates who fall below the threshold would receive no delegates even if the threshold were removed, and the paradox occurs because of some rearranging of the delegates among the surviving candidates. Of the 21 elimination paradoxes in the data, only the Iowa caucuses at-large example exhibited the elimination paradox when an eliminated candidate had earned delegates with no threshold. The 2020 Iowa caucuses provided the first known example of such an extreme instance of this paradox ever to occur in a Democratic presidential primary, and it occurred at Biden's expense.

Alabama Paradox

Suppose that instead of nine at-large delegates to apportion, there were eight. How should that affect the at-large apportionment of (3, 3, 2, 1)? The reasonable answer is that one candidate should lose a delegate, but otherwise, the apportionment should remain the same. Perhaps the apportionment should change to (3, 2, 2, 1), for example. However, if we decrease the number of at-large delegates from 9 to 8 and run Hamilton's method (still with the 15% threshold in place), the apportionment is (2, 2, 2, 2), and Biden gains a delegate. Table 4 shows how this occurs: when there are eight delegates, Biden's fractional part is the second largest, earning him an extra delegate, but when there are nine delegates, Biden's fractional part is the smallest. Biden would have been better off with fewer at-large delegates to apportion: increasing the number of delegates from 8 to 9 cost him a delegate.

This outcome is an example of the *Alabama paradox*, so named because it was first discovered in 1880 when a census clerk noticed that under Hamilton's method, if there were 299 seats in the US House, then Alabama would receive eight seats, but if the 299 were increased to 300, then Alabama's apportionment would decrease to seven. The paradox occurs when an increase (respectively decrease) in the number of seats causes the apportionment for a state to go down (respectively up). When this paradox was pointed out in 1880, it was merely a mathematical curiosity; however, it became an enormous political problem 20 years later, and arguably this paradox is why Hamilton's method is no longer considered a viable method for apportioning House seats in the United States (see [1]).

After the 1900 census, Congress needed to agree on the size of the House,[8] and the House Select Committee on the Twelfth Census submitted a report containing Hamilton apportionments for all House sizes between 350 and 400. As the size of the House increased from from 350 to 400 one seat at a time, the apportionment for the state of Maine oscillated between three and four seats. The Committee proposed a bill that would fix the size of the House at 357, one of the House sizes that gave Maine only three seats. During the debate over this bill, John Littlefield, a representative from Maine, gave my favorite quotation ever uttered by a politician about the mathematics of apportionment:

> Maine loses on 382. She loses again with 386, and does not lose with 387 or 388. Then she loses again on 389 and 390 … [I]t does seem as though mathematics and science had combined to make a shuttlecock and battledore[9] of the state of Maine in connection with the scientific basis upon which this bill is presented … In Maine comes and out Maine goes … God help the state of Maine when mathematics reach for her and undertake to strike her down!

As with the elimination paradox, divisor apportionment methods are not susceptible to the Alabama paradox (this is quickly proven given the sequential nature of the methods), which is partially why such methods have been used for House seat apportionment (in the United States and elsewhere) since the early twentieth century. This could be another point in these methods' favor in comparison to Hamilton's method in the context of delegate apportionment.

The Alabama paradox is rarely observed in presidential primaries. In the 2,206 elections I investigated in my Democratic database of primary elections,[10] I found only one occurrence of this paradox: the 2020 apportionment of the at-large delegates in the Iowa caucuses. This likely means that the 2020 Iowa caucuses provided the first example of this paradox ever to occur in a Democratic presidential primary.[11] This occurrence of the Alabama paradox, like the at-large occurrence of the elimination paradox, comes at the expense of Biden.

[8] The size of the House has been fixed at 435 by law since 1913. Prior to 1913, Congress would often change the size of the House after a census to keep pace with an increasing population.

[9] This was the small racket used in early versions of badminton.

[10] I am able to use more elections here because, in contrast to the elimination paradox, I don't need vote information for eliminated candidates to see whether the Alabama paradox occurred.

[11] One of the reasons this paradox occurs so rarely is that the Democrats use a 15% threshold, which produces elections with fewer candidates, which in turn decreases the likelihood of this paradox being observed. If we go through the Democratic database of elections and look for instances of the paradox when no threshold is used, we find 83 instances of the paradox (including the Iowa at-large example), which is still relatively small. As we increase the threshold from 0% to 15%, the number of instances of the Alabama paradox steadily declines.

Population Paradox

Note from Table 3 that the SDE results for Buttigieg, Sanders, Warren, and Biden were $\mathbf{s} = (164.2389, 159.2701, 119.0656, 92.8112)$ in CD 3, resulting in an apportionment of $(3, 2, 2, 1)$. Suppose instead that the SDE results were $\mathbf{s}' = (165, 164.7, 164.5, 91.8)$. Should $\mathbf{s}'$ produce better or worse results for Biden? The reasonable answer is that Biden should prefer $\mathbf{s}$ to $\mathbf{s}'$: in moving from $\mathbf{s}$ to $\mathbf{s}'$ he has lost SDEs, whereas every other candidate gained SDEs. Sanders and Warren in particular have put substantially more distance between themselves and Biden. When I pose questions like this to students and ask how many delegates Biden's 91.8 SDEs should earn him from $\mathbf{s}'$, they answer either 0 or 1. However, the Hamilton apportionment for $\mathbf{s}'$ with the 15% threshold is $(2, 2, 2, 2)$. This illustrates an example of the *population paradox*, whereby there exists a second hypothetical election result in which one candidate's vote/SDE count goes up (Buttigieg) while another's goes down (Biden), yet the former candidate loses a delegate to the latter. In CD 3, Biden was hurt in some way by having a higher SDE count; in an alternate universe, he does better with fewer SDEs, while Buttigieg (and the other two candidates) have more. If only Biden had campaigned for Sanders and Warren in CD 3 (while neglecting his own campaign a little bit), perhaps he could have gained an extra delegate in this district.

How often does the population paradox occur in practice? Since the paradox depends on having two different vote distributions, the answer depends on what we mean by "occur." One way the paradox could actually occur is if a recount of SDEs occurred in CD 3, causing the initial distribution $\mathbf{s}$ to change to $\mathbf{s}'$. This kind of drastic change in SDEs after a recount is incredibly unlikely (and there was no recount in CD 3), so instead of using the language of "occurrence," many social choice theorists instead speak of a given vote/SDE distribution $\mathbf{v}$ as being *susceptible* to the paradox. In this framework, $\mathbf{v}$ is susceptible to the population paradox if there exists another distribution $\mathbf{v}'$ such that in moving from $\mathbf{v}$ to $\mathbf{v}'$, one candidate's vote total increases while another's decreases, yet the former (respectively latter) candidate loses (respectively gains) a delegate. Unfortunately, I am unaware of necessary and sufficient conditions to check whether a given vote distribution is susceptible to the paradox, and thus there is no meaningful way for me to query my Democratic primaries database to check for frequency of susceptible elections. However, in [4], Brent Bradberry visualizes the elections that are susceptible to this paradox in three-candidate elections using simplicial geometry, and we can see that the vast majority of elections are not susceptible. Therefore, we have some evidence that susceptibility to the population paradox is a rare occurrence in elections with a small number of candidates, and thus the susceptibility of the SDE distribution in CD 3 is probably not common.

The population paradox never featured prominently in political debates over apportionment in the United States[12] (there were no representatives screeching about mathematics striking down their states in reference to this paradox, for example), but avoiding this paradox drove much of the mathematical study of apportionment in the twentieth century. As Michel L. Balinski and H. Peyton Young state in their seminal text [1], "The population paradox violates the fundamental idea that changes in apportionments ought to reflect correctly changes in population." Replace "population" with "votes," and the same sentiment applies to delegate apportionment. Balinksi and Young proved that divisor methods are precisely the methods that are not susceptible to this paradox, and as a result, their famous impossibility theorem, which states that no apportionment method can both satisfy the quota rule and avoid the population paradox, is featured in many Math for the Liberal Arts-type textbooks. Unfortunately for Biden, the DNC (presumably) did not consult [1] when they were selecting an apportionment method.

Aggregation Paradoxes

The Iowa caucuses constitute a statewide election to determine how many of the Iowa Democratic Party's delegates each candidate will receive. But instead of apportioning delegates in a single calculation based on the statewide performance of the candidates, the Iowa Democratic Party apportions delegates in six separate calculations and then sums the results to determine each candidate's apportionment, as mandated by the DNC. Table 3 shows that Biden received six delegates in total as a result of summing over these six separate apportionments. However, if all 41 delegates had been allocated in a single statewide Hamilton calculation, the apportionment would have been $(12, 12, 9, 8, 0)$ for the top five candidates instead of the $(14, 12, 8, 6, 1)$ from Table 3, and Biden would have received eight delegates. Paradoxically, for Biden, the sum of the parts falls well short of the whole. If Biden's 15.9% of the statewide SDEs earns him eight of 41 statewide delegates, why should he lose two of those delegates when the single apportionment problem is broken into six subproblems? This outcome is an example of an *aggregation paradox*, which was first studied in [9]. We say that this paradox has occurred when the apportionment obtained from breaking the delegates into several categories and summing over those separate apportionments is different from an apportionment obtained from a single statewide calculation. This kind of paradox is reminiscent of what can occur in the Electoral College, when a candidate can win the popular vote, yet when votes are translated into electoral votes and then aggregated across the Electoral College, the candidate loses the presidency.

[12] Internationally, the population paradox has received a bit more attention. The German Bundestag discarded Hamilton's method because of its susceptibility to this paradox, for example, and adopted a method that does not suffer from it (see [13]).

Perhaps you don't think that this example rises to the level of the term *paradox*. After all, across the six apportionment problems we use five different SDE distributions, so maybe it is expected that the statewide apportionment will not equal the sum of the parts when we sum apportionments in this manner. Even though I defend the use of the term "paradox" here, the data does bear out that this kind of outcome is to be expected. In my database of Democratic elections there were 183 state primaries for which I had enough CD data to check whether the aggregation paradox occurred when taking into account the CD, PLEO, and at-large apportionments. This paradox happened in 115 elections, well more than half of the state primaries in the database.

The Iowa caucuses demonstrate another kind of aggregation paradox, though, that maybe you will think deserves the term "paradox." Recall that the five PLEO and nine at-large delegates are apportioned separately, but each apportionment is based on the exact same underlying SDE distribution. Using two separate calculations in the actual apportionment earned Biden a total of two statewide delegates, one PLEO and one at-large. However, if we allocate the 14 statewide delegates in a single calculation, then the apportionment is (4, 4, 3, 3), and Biden earns an extra delegate. In other words, Biden lost a delegate because the DNC decided that the PLEO and at-large delegates must be allocated separately, even though they are both apportioned based on the same statewide SDE data. This is a more egregious example of an aggregation paradox, because the division of PLEO and at-large delegates seems completely arbitrary; in our earlier aggregation example, perhaps it makes sense to aggregate over CDs if some candidates have poor statewide performances but strong performances in some CDs. But in this case there seems to be no purpose, either mathematically or politically, for allocating the PLEO and at-large delegates in two separate calculations.

This PLEO/at-large version of the aggregation paradox[13] is a rarer occurrence than the aggregation paradox that occurs when we take CDs into account. I checked the 210 state primaries[14] in my Democratic primary database that contained both PLEO and at-large delegates, and this kind of aggregation paradox occurred 58 times, much less frequently than when we also take the CD results into account.

Divisor apportionment methods are immune to the previous paradoxes, but even they succumb to aggregation paradoxes. If Democrats had used Jefferson's (respectively Huntington–Hill's) method in their primaries, then the Democratic database of elections would have returned 137 (respectively 122) instances of the CD version of the aggregation paradox. For the PLEO/at-large version, the number of instances of the paradox for elections in the database is 64 for Jefferson's method and 59 for Huntington–Hill. These numbers are larger than the corresponding numbers for Hamilton's method, so perhaps we have found a small point in favor of Hamilton.

We close this section by noting that the issues of aggregation also have some conceptual relationship to the Alabama paradox. One way to think about the Alabama paradox is that small perturbations in the number of delegates should produce small and reasonable changes in the apportionment. Suppose that in the Iowa caucuses we keep the overall number of delegates equal to 41, but shuffle them around a little bit by sending one at-large and one CD 2 delegate to CD 4, so that now CD 4 has seven delegates to apportion and there are six (respectively eight) delegates in CD 2 (respectively at-large). You can check that this small perturbation of delegates increases Biden's statewide apportionment by two, earning him eight total delegates. That is, Biden could have earned 33% more delegates if only the delegates to be apportioned had been slightly shuffled within the six delegate categories.

Conclusion

The 2020 Iowa Democratic caucuses exhibited a historic amount of apportionment paradoxes, all of which worked against Joe Biden. He eventually won the Democratic primary and subsequently the presidency, so I doubt he would be particularly bothered by the harm he suffered as a result of these paradoxes. On the other hand, it would be completely understandable if, in a fit of exasperated indignation after waiting so long for the results of the caucuses, Biden had clenched his fists and howled at the DNC, "God help the Biden campaign when mathematics reach for her and undertake to strike her down!"

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David McCune, Department of Mathematics and Data Science, William Jewell College, 500 College Hill, Liberty, MO 64068-1896, USA. E-mail: mccuned@william.jewell.edu